JOINT CALL 2023

PROJECT DESCRIPTION TEMPLATE

FULL PROPOSAL

Proposal code	CETP-2023-00138
Project Acronym	TenSyGrid
Project Title	Tensors for System Analysis of Converter-dominated Power Grids
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1. EXCELLENCE

1.1 Overall aim and objectives

Tensors for System Analysis of Converter-dominated Power Grids (TenSyGrid) will develop a toolbox with innovative stability analysis methods to enable and facilitate the integration of 100 % renewable-based energy systems while maintaining a safe and stable operation of the power system. The nature of the power system is radically changing due to the massive deployment of power electronics required for, among others, renewables integration into the grid. Traditional assumptions to simplify power system simulations, e.g., root means square (RMS) dynamic simulations, fail to predict fast transient behaviors when networks are dominated by converters [1], i.e., have a High Penetration of Power Electronic-interfaced Power Sources (HPoPEIPS). As a result, there are uncertainties on the new power system dynamics of low-inertia converter-dominated networks. These uncertainties essentially limit the integration of renewable energy sources. For example, the Irish transmission system operator EirGrid currently limits the instantaneous penetration of variable renewable energy (VRE) to 75 % [2]. These conservative limits are needed as a precautionary measure to ensure a stable operation. System operators need reliable tools during, for example, offline grid integration studies of renewable power plants, or HVDC links, as well as tools that support them in the root-cause analysis during real-time operation of networks with HPoPEIPS. Therefore, TenSyGrid has the following main objective:

Main Objective: Developing a toolbox, called TenSyGrid Toolbox, with innovative direct-stability assessment methods to support decision makers, mainly network operators, during the real-time operation of large-scale power networks with 100 % renewable energies.

Thereby, TenSyGrid addresses the R&I Theme of “large-scale renewable generation and system flexibility and reliability” and “system integration and flexible operations”. Basically, this means that the electricity network is the central infrastructure to integrate different energy forms, such as hydrogen or process heat, while being powered by VRE. This paves the way for an integrated energy system with zero emissions (TRI1). TenSyGrid follows five specific objectives (SOs), depicted in Figure *1*, that are motivated from the state-of-the-art approaches in power systems and the challenges they are facing for allowing 100% renewable energy-based networks in different regions of the world.

SO1: Obtaining a modeling framework for power systems with high penetration of power electronic devices that captures the relevant nonlinear dynamics.

There is a need for a modeling framework that is simpler than general nonlinear modeling but that captures the relevant nonlinear power system dynamics, leading to a power system component model library. As renewable energy penetration increases, an accurate understanding of the aforementioned nonlinearities may become critical for ensuring a stable operation.

SO2: Assist decision makers with real-time power system stability assessments under high penetration of renewable energy.

Better stability assessment methods are required for networks with HPoPEIPS that can support root-cause analysis during real-time operation. Ultimately, such methods could improve the safety margins of system operators, enabling an even higher level of renewables penetration.

SO3: Overcome the lack of interpretability of black-box models from original equipment manufacturers hindering root-cause analysis.

There is a lack of system identification methods for original equipment manufacturer (OEM) black-box models that can obtain interpretable models. Moreover, obtaining interpretable models that go beyond linear systems can truly open the door for more advanced root-cause analysis.

SO4: Obtaining a modeling framework that is easily scalable to extremely large power systems and easily updatable to address ongoing structural changes in the power system network.

An easily scalable and updatable model class is essential to address the ongoing structural upgrades and changes the power system is subject to. The power system is ever changing; thus, any static modeling framework is doomed to rapidly become outdated.

SO5: Developing tools that are compatible with existing commercial software packages and new methodologies that are easily accessible.

Lastly, it is undeniable that there have been many efforts in different projects on designing new modeling frameworks for energy systems. Crucial for the actual impact of new methodologies and modeling frameworks is the compatibility with existing commercial software packages to reduce the need of daily practitioners to familiarize themselves and adapt to new workflows.

The objectives of TenSyGrid are very ambitious. It addresses the lack of analytical tools in daily practices of transmission and distribution system operators (TSOs & DSOs) for networks with HPoPEIPS. By that, TenSyGrid will enable integrating more renewables by reducing conservative margins of the integration limits.

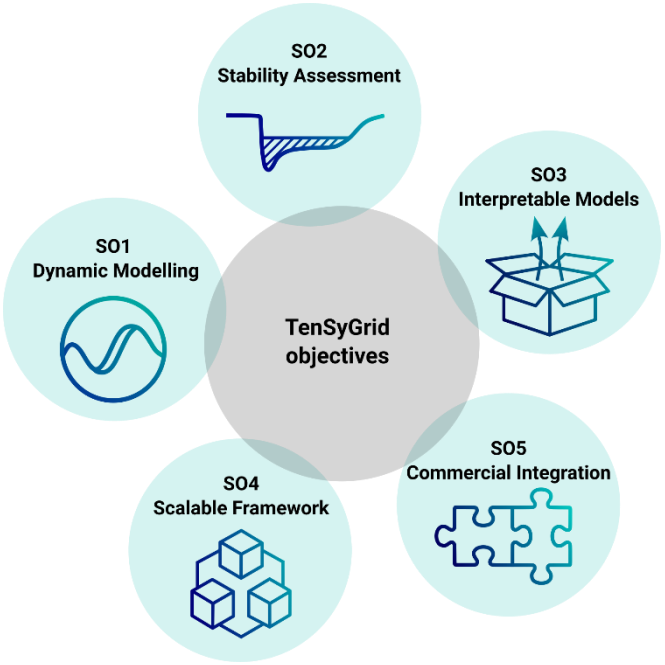


Figure 1: Representation of TenSyGrid's objectives

1.2 Pertinence of the objectives to the Call and Call Module

The TenSyGrid project, with its focus on advancing the state of power system analysis through the development of innovative direct-stability assessment methods and a comprehensive modeling framework, directly aligns with the European Union's ambitious goals for a sustainable and green energy future. The project's objectives are particularly pertinent to the call, as they address the pressing need for reliable stability assessment methods in power grids with HPoPEIPS, a challenge that becomes increasingly relevant as Europe moves towards 100% renewable energy sources.

The pertinence of TenSyGrid's objectives to the call module 2 "Energy system flexibility: renewables

production, storage and system integration" is underscored by its direct contribution to enhancing grid stability, enabling the integration of more renewables, and reducing conservative margins of integration limits. As we understand, model, and analyze the system better with an improved mathematical framework that better captures the power electronic (PE) fundamental behavior, we can introduce more VRE. This fully aligns to the objective of the Innovation Priority (IP) 2.3.1 "System stability assessment considering high VRE penetration" of the Green Powered Future Mission (GPFM) [2]. This IP scored the highest importance in the survey for the multilateral research program. TenSyGrid not only tackles the technical challenges associated with HPoPEIPS but also proposes innovative solutions that have the potential to revolutionize the way power system stability is assessed and managed. By focusing on overcoming the lack of interpretability of black-box models from OEMs, TenSyGrid addresses a critical barrier to effective root-cause analysis and real-time operation support for system operators.

Furthermore, TenSyGrid's ambition to develop a modeling framework that is scalable, easily updatable, and compatible with existing commercial software packages emphasizes its relevance to the call's objectives. This approach ensures that the project's outcomes will be directly applicable and beneficial to the daily practices of transmission and distribution system operators (TSOs & DSOs), ultimately facilitating a more seamless integration of renewable energy sources into the grid.

1.3 Project ambitions

1.3.1 State-of-the-Art Power System Analysis with 100 % Renewable Energies

Power systems worldwide are undergoing a fundamental change towards HPoPEIPS powered by renewables that introduce complex dynamics, and thereby, challenges the state-of-the-art practices of power system analysis. The power converters of renewables present a programmable behavior that can interact in wide range of frequencies (from Hz to kHz), which might lead to potential interactions between devices. Power system analysis of large-scale networks is typically done by RMS simulations. One major problem for RMS models is that these interactions of the fast converter controllers with each other and other network elements, for example HVDC converter with other converter-connected devices, could lead to instabilities. RMS models may not reliably predict these instabilities and therefore provide an exceedingly optimistic behavior. As a solution, more detailed electro-magnetic transient (EMT) models are used. Their computational burden becomes so high, however, that EMT simulations of large-scale networks are currently not practical [2]. This motivates SO1.

A general challenge of modeling power systems is that they are hybrid nonlinear systems, i.e., have continuous and discrete dynamics. Converters introduce many nonlinearities, for example, switching, current and voltage saturations, products of voltage times current, or the phase-locked loop of angle imposition [3]. This hybrid nonlinear nature of the power network and its components raises significant challenges for the necessary modernization of state-of-the-art control concepts to ensure stable closed-loop operation. The fast converter controllers may interact with each other as well as other power system components, potentially leading to system instabilities. A standard approach to assess power system stability is to linearize the system and then perform an eigenvalue analysis to identify critical interactions. However, the small-signal regime analysis breaks down fast when network disturbances become larger. This case occurs when transitioning to low-inertia networks with HPoPEIPS where, e.g., rate of change of frequencies (ROCOF) increase, or short-circuit faults become critical. In such situations the system is no longer in a small signal regime and the linear model might lose accuracy. System operators are therefore only left with time-consuming EMT studies to analyze such detailed nonlinear interactions. These reasons motivate SO1.

Typically, system operators repetitively run a set of static and dynamic simulation analysis during real-time operation to identify critical scenarios, e.g., power flow, contingency analysis, or short circuit current analysis. Computational speed is therefore critical, which is why simplified RMS, or linearized models, are used. Contrary to time-domain simulation, that goes through one scenario after the other, analytical stability assessments provide a general picture of the system dynamics, e.g., providing the stability margins. Thereby direct stability assessment methods are very powerful. However, comprehensive approaches on Lyapunov functions, e.g., [4] [5], have not been able to provide an alternative to time-domain simulations for system operators. That is because it is generally challenging to find Lyapunov candidates, and it typically requires simplifying assumptions. Nevertheless, other direct stability assessment methods like equal area criterion may help system operators to reduce the number of critical cases to be frequently simulated during operation. Yet, such methods are designed to networks dominated by synchronous generators and not low-inertia networks with HPoPEIPS. The gap in the current state-of-the-art approaches motivates SO2.

Another challenge is the construction of network models for modern power systems due to the intellectual

property restrictions imposed by OEMs. For new installations, system operators typically receive black-box models developed by the converter manufacturers, where limited access to only high-level control layers is granted. These restrictions challenge power system engineers to identify the source of instabilities using commercial tools and rely on their experience for the identification. To assess stability with such models, power engineers have developed tools that do not require access to the internal control structure, such as impedance measurement and analysis tools [6]. Using frequency excitation sweeps, such tools are capable of computing impedance models of both the converter under analysis and the power network, to later assess stability using, e.g., the generalized Nyquist criterion, or other alternatives. However, despite the relevance of such an approach, the obtained models are only valid during small disturbance regimes. Having interpretable nonlinear models would greatly benefit the stability assessment. The need for interpretable models of black-box models of OEMs motivates SO3.

The need to meet the growing electricity demand due to the electrification of other sectors by renewable energy sources will lead to dramatic increase in the complexity and number of devices. Therefore, modeling frameworks need to be easily scalable. Current approaches rely on, e.g., finding an equivalent representation of aggregated subsystems, or investigating hybrid simulation approaches that combine RMS and EMT-models [1]. Additionally, the network is subject to continuous structural changes, meaning that models can easily be outdated. The need for ease of scalable and updateable models motivates SO5.

1.3.2 Hybrid Implicit Multilinear Modeling for Power System Analysis

Multilinear time-invariant (MTI) models have been developed by our full partners at Hamburg University of Applied Sciences (HAW) and the Fraunhofer Institute for Wind Energy Systems (IWES) to be capable of capturing more complex dynamic phenomena than linear time-invariant (LTI) models. The term multilinear describes a function of several variables, discrete or continuous, which becomes linear if all except one variable are held constant, e.g., $f(x,y) = a_0 + a_1x + a_2y + a_3xy$ (notice that a term with, e.g., x^2, would not be allowed). This differs from multi-linear or piecewise-linear functions, which are linear in predefined regions [7]. In general, multilinear functions are a richer subclass of nonlinear functions, which includes linear functions and more interestingly binary functions, see Figure 2 (a).

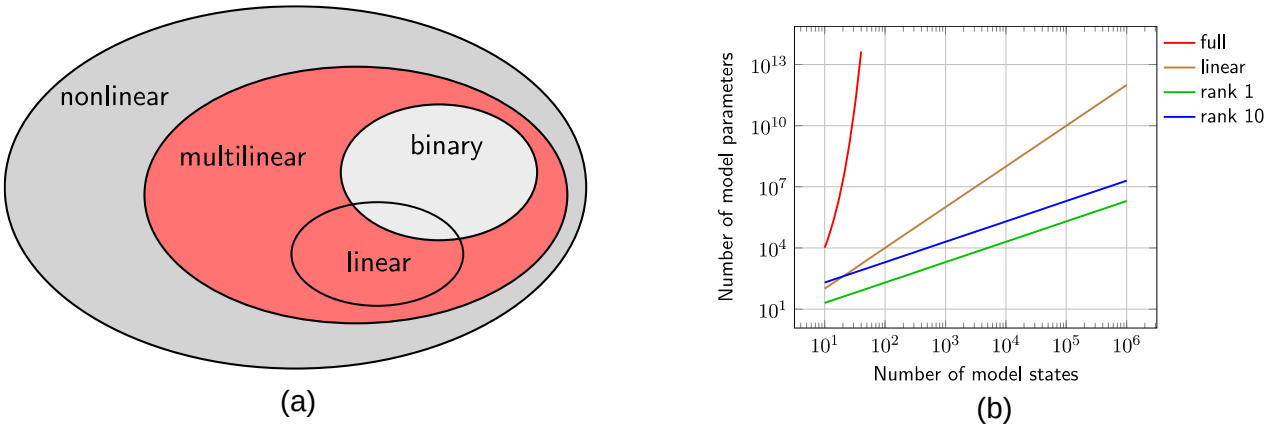


Figure 2: (a) Nonlinear model space subclasses. (b) Model parameters vs states and rank.

Moreover, due to said multilinear structure, MTI models can be represented as tensors - like LTI models as matrices of state space parameters. A tensor generalizes the notion of a matrix as a two-way array of numerical values to a multi-way array. Tensors can be represented very compactly via tensor decomposition methods like canonical polyadic (CP) [8] decomposition, which enables performant computations of high-dimensional models [9], breaking the curse of dimensionality. For example, assume a measurement signal of 2 years sampled every minute, which amounts to around 10^6 data points. All the measurement data points can be stored in a 4-dimensional tensor as shown in Figure 3 (a). Notice that the choice of dimension relates to interpretable periodicity concepts, which ensures a meaningful location for each data point with respects to its neighbors, e.g., day 3 (Wednesday) of week 5 is adjacent to day 3 of the weeks 4 and 6 for the same year. Using tensor decomposition methods, a rank 1 approximation depending only on the size of each dimension can be identified, which only requires around 1500 data points, around a factor 10^3 of compression, see Figure 3 (b). Naturally, if the behavior of the full tensor is rather complex, more than one rank 1 decomposed tensor might be required to accurately approximate it. However, as the number of data points required to represent a full tensor increases exponentially with the number of dimensions, it only increases linearly for decomposed tensor with a slope given by its rank. While the example was concerned with data points in the signal space, the same applies to parameters in the model space (the focus of MTI),

as seen in Figure 2 (b).

The technique of tensor decomposition has been used on parameter spaces in, for example [7], and [11]. The first MTI Toolbox 1.0 available for Matlab was published in 2017 for the explicit multilinear time-invariant (eMTI) modeling class [12]. The explicit multilinear class and the MTI Toolbox has been used for different applications, like heating systems [7], building automation [13], or modeling of an electrolyzer [14]. An important recent development has been the introduction of the implicit multilinear modeling class [10], because it extends the capability of modeling nonlinear dynamics. A special format of tensor decomposition that allows reducing the required storage by almost half was introduced in [13]. The specialized format was utilized in [15] to develop an efficient linearization algorithm of explicit multilinear models. Therefore, TenSyGrid builds upon recent developments and an already existing toolbox.

In the context of power systems, MTI models, in particular hybrid implicit MTI (iMTI) models [10], are able to capture the most relevant nonlinearities, e.g., switching states, converter saturations (through Boolean variables and inequalities), or trigonometric functions in the Park transformation (via system order increment and auxiliary variables definition for trigonometric functions). For example, Figure 4 shows a comparison of a semi-explicit multilinear phase-locked loop model against a nonlinear trigonometric counterpart. The example consists of a converter connected to an equivalent grid representation through a filter and a line section. The plots show that even under forcefully induced unstable operating conditions, the multilinear model is able to depict the same behavior as the nonlinear one. This scenario is particularly relevant, as it is precisely when systems are driven away from their standard operating conditions that nonlinearities play a major role.

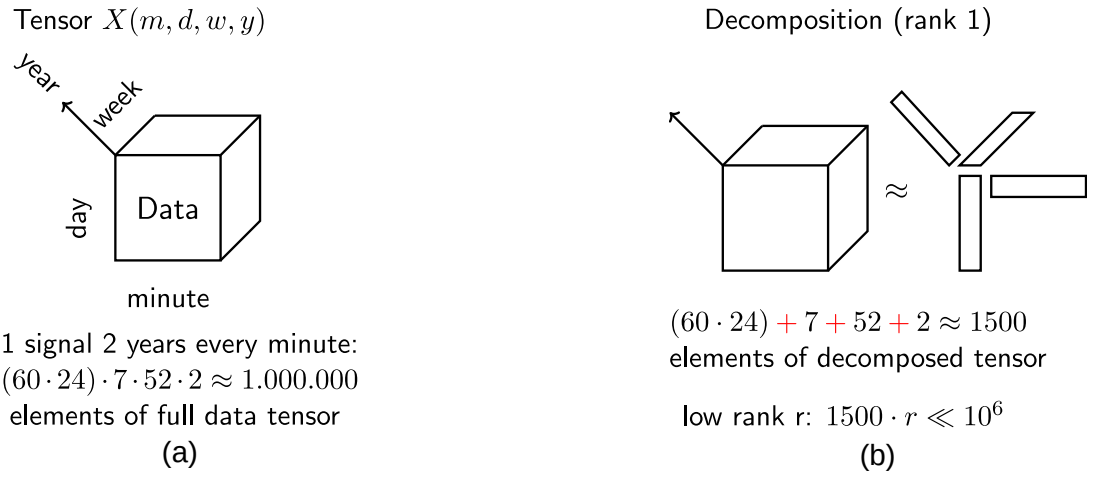


Figure 3: (a) Full data tensor representation. (b) Rank 1 decomposed tensor representation.

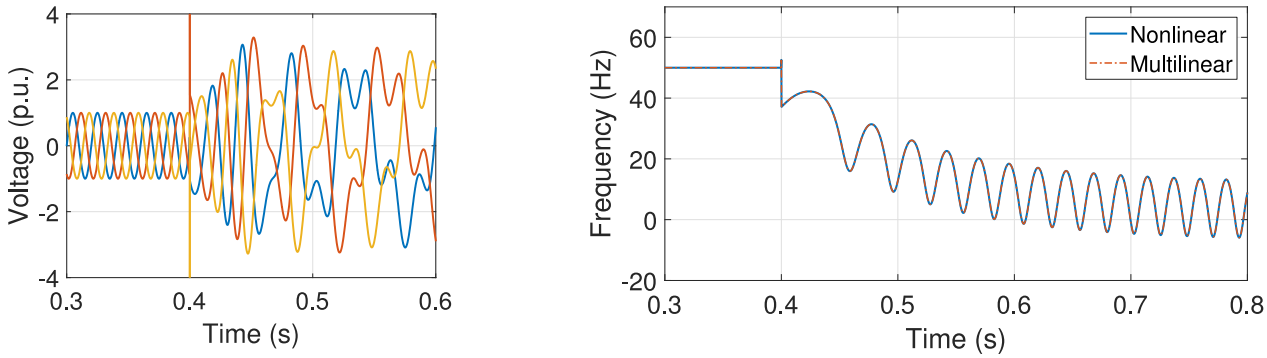


Figure 4: (a) Grid voltage. (b) Comparison of the nonlinear and the multilinear model of the frequency estimation by the phase-locked loop under the induced unstable operating conditions triggered current infeed of converter at 0.4 seconds.

1.3.3 Expected Impact on Power System Analysis with 100 % Renewables

Departing from a TRL of 3, TenSyGrid extends the hybrid implicit multilinear modeling class and improves

the MTI Toolbox with new functionalities that are required for the TenSyGrid Toolbox. Therefore, first some important power system components are taken, to identify the nonlinearities that need to be captured by the hybrid iMTI models. Then suitable modeling approaches can be developed for the iMTI class, e.g., specific change of variables, or multilinearisation techniques. Once an understanding of what are the dynamics required and how they are represented by hybrid iMTI models, suitable tensor decomposition techniques can be explored to find a minimal format. Meanwhile, multilinear models of several power system components can be developed as well as the multilinear system identification techniques. With the hybrid iMTI models being in a specific structure, stability assessment methods can be explored and developed that utilize the decomposed tensor structure. With the library of power system components, multilinear system identification, and stability assessment methods, the TenSyGrid Toolbox can be developed into a framework that is relevant for power system engineers and potentially compatible with commercial software. During this process, the capabilities will be tailored to the needs of power systems, as the example networks grow from small to very large system to harness the potential of tensor decomposition for power system analysis. Finally, these developments will be tested on real-world networks and benchmarked with commercial tools to demonstrate that a TRL of 5 has been achieved by the end of the project. Furthermore, the viability for a potential commercialization project will be assessed.
The ambitious goal of enabling direct-stability assessments for large-scale networks with 100% renewable energy highlights TenSyGrid's commitment to contributing to Europe's energy transition towards a more sustainable future.

Moreover, the project's emphasis on ensuring the practical application and deployment of its outcomes highlights its holistic approach. By engaging with stakeholders, validating methodologies through real-world applications, and focusing on user accessibility, TenSyGrid aims to bridge the gap between advanced research and practical utility. The project's ambitions are reflective of a deep understanding of the systemic changes needed within the power system landscape to accommodate the increasing share of renewable energy sources, demonstrating its relevance and potential impact on Europe's energy sector. The project goes way beyond the state-of-the-art as traditional simulation approaches, which are still predominant these days, rely on solving a set of non-linear equations with great computational effort. The transformation of these equations into multilinear systems, together with the concept of tensor decomposition, can make the solution of TenSyGrid various orders of magnitude faster than current technologies. It has the potential to reach a TRL of 5 at the end of the project.

1.3.4 Challenges on the Implementation of Hybrid iMTI for Power System Analysis

There are some challenges to the goals of the projects. A first challenge in the development is to ensure that the chosen modeling class truly captures the relevant nonlinearities, particularly when decomposed to lower rank approximations. To improve the chances of success, TenSyGrid is opting for a rich closed modeling class like hybrid iMTI, which already has showed to capture some of the relevant dynamics as mentioned before. Moreover, the expertise of our power systems partners UPC, eRoots, and UM, will aid on the formulation of a robust validation process to thoroughly test the developed models.
An important challenge of the project ambition is achieving meaningful stability assessment based on lower rank hybrid iMTI models that go beyond eigenvalue analysis. The expertise of the partners HAW, IWES, and the UPC will be utilized to obtain fast system analysis methods. Together with the associated partners, like the Technical University of Delft, or the French TSO RTE, feedback loops are embedded to work on the right assessment of power system stability.
As with any new toolbox, one of the main challenges is the adoption by its end users. To address this challenge, the TenSyGrid toolbox will be developed in close collaboration with industry partners and stakeholders in the advisory board. Moreover, TenSyGrid is opting for an open-source strategy which provides transparency for its user, i.e., the end-user will have access to the fundamental inner workings of the toolbox so practitioners can better understand their obtained results.
As part of its exploitation strategy, TenSyGrid is planning interfacing with existing commercial software. The idea is to have a complementary role and not to replace existing tools altogether. However, this naturally poses a challenge, as there are no universal standards for such power system software solutions. Therefore, as part of its development, TenSyGrid aims at exploring different integration opportunities with existing commercial software solutions under the guidance of our partner eRoots and its experience in the utilization and development of power system software.

1.4 Methodologies developed or applied to achieve the project objectives

TenSyGrid will develop a set of analysis methods which will be bundled in the TenSyGrid Toolbox with the

multilinear modeling framework as its foundation, see Figure 5. TenSyGrid will foster synergies between the power systems and control systems communities. The development of the TenSyGrid toolbox follows the SOs as central guidelines for its implementation.

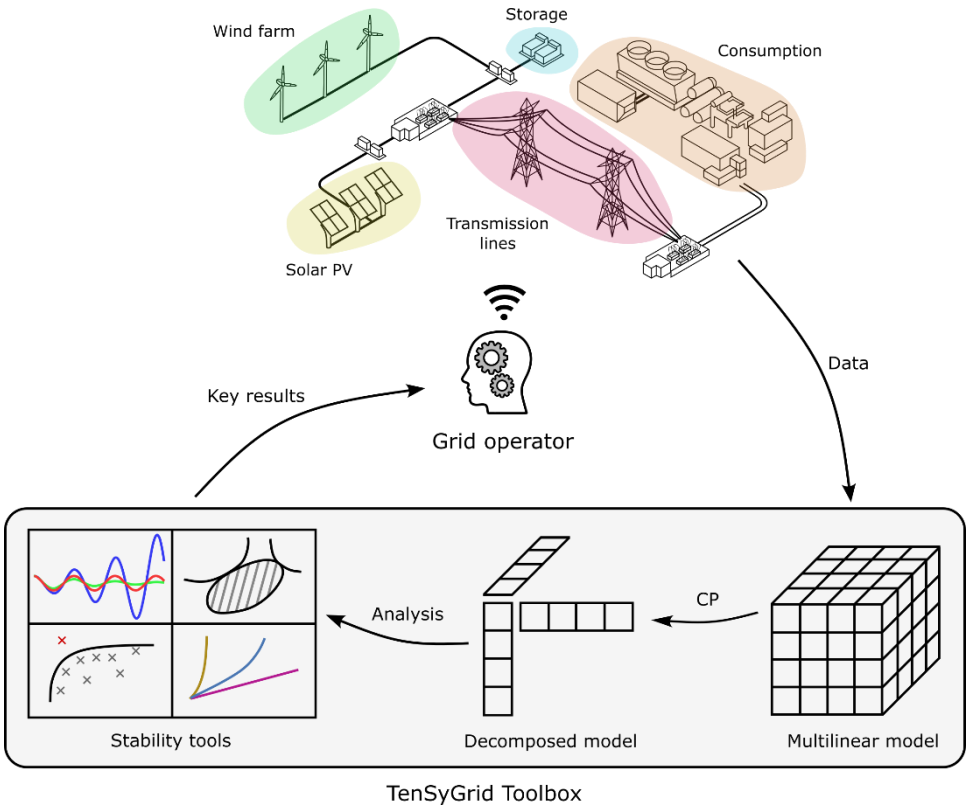


Figure 5: Overview of the TenSyGrid Toolbox and connection with the grid operator.

SO1: Obtaining a modeling framework for power systems with high penetration of power electronic devices that captures the relevant nonlinear dynamics.

The key aspect that limits the applicability of current linear analysis approaches is the future integration of 100% renewable energy in the power system. As mentioned in the previously, when the amount of converted-interfaced renewable energy sources increases, the nonlinearities introduced by the involved converters and their interactions become more prominent, particularly due to the lack of inertia in the system. In this context, hybrid iMTI, can capture the most relevant nonlinearities of power systems, as mentioned in previous subsections. Thus, hybrid iMTI is a great candidate modeling class for a power system component library.

SO2: Assist decision makers with real-time power system stability assessments under high penetration of renewable energy.

Under converter-dominated nonlinear regimes, power system stability assessment becomes really challenging, in particular for real-time assistance. Here, the classic scenario-based approaches are too computationally heavy to meet the real-time demands. One possible solution is an eigenvalue analysis based on the linearization of nonlinear models, where recent results indicate that extremely efficient linearization of MTI models are possible [16]. For large signal analysis, more comprehensive solutions, like the computation of region of attractions are also possible, e.g. by cell mapping, [17]. However, these methods suffer from the curse of dimensionality, which can be tackled by tensor decompositions. First decomposition results for this approach are shown in the field of qualitative modelling by stochastic automata, which also belongs to the class of MTI models [18]. Alternatively, methods developed by TenSyGrid could provide support on the identification of critical scenarios for time-domain simulations by discarding subsets which do not cause major stability concern given certain boundary conditions. Contrary to equal area criteria, which were designed for synchronous generators, such method will be able to handle a wider range of nonlinear dynamics through the hybrid iMTI class, potentially leading to less conservative, but validated decisions. Moreover, we could potentially avoid scenario-based analysis by simulations, as shown next.

TenSyGrid will employ an approach to assess stability in multilinear systems without the need for simulations. This is comparable to linear systems stability analysis, which involves examining the coefficients of a specific matrix decomposition. However, contrary to the linear case where stability is a property of the system, for the

multilinear case (as for general nonlinear systems) it depends on the solution analyzed (e.g., fixed point, periodic). In a similar fashion to the matrix decomposition notion of the linear system case, we formulate a tensor decomposition for the multilinear case. This decomposition is exactly what will be estimated by the system identification algorithms developed in this project. Unique canonical polyadic (CP) representations are good but not exclusive candidates for decompositions methods which support stability analysis for MTI systems [13]. Restricting our analysis to decomposed MTI models, which already capture the most relevant power systems nonlinearities, increases the chances of finding high performance analysis methods for this special subclass that can represent the main power system stability phenomena, unlike general nonlinear methods.

SO3: Overcome the lack of interpretability of black-box models from original equipment manufacturers hindering root-cause analysis.

The MTI class offers explainable models – in contrast to standard artificial intelligence (AI) and machine learning (ML) methods based on artificial neural networks (ANNs) typically used for black-box models. This can be seen in particular when using decomposed tensor formats like the norm-1 CP decomposition [19] (norm-1 CPN), where influence factors and their contribution over time can be interpreted in natural language, e.g., “the influence of x on y has risen in the last z minutes by a critical factor of 2”. The aim is to develop new system identification methods tailored to decomposed MTI models parting from the existing expertise in more general nonlinear systems.

SO4: Obtaining a modeling framework that is easily scalable to extremely large power systems and easily updatable to address ongoing structural changes in the power system network.

An aspect that always restricts the applicability of any method for power systems, particularly general nonlinear approaches, is scalability. Just by considering a few buses in a model, the number of states and in particular the number of possible connections and configurations when considering hybrid models skyrockets really fast. The advantages of MTI models, besides natively supporting hybrid configurations, are their tensor format and the available tensor decomposition methods. The key insight here comes from the tensor rank. The more structured a tensor is (due to symmetries, identical components, easy connectivity structures like parallel or series), the lower the rank needed to make important assessments about the system, e.g., the largest eigenvalues of the linearized system matrix or the ROCOF. Different tensor decompositions can be explored beyond CP, like tensor trains (TT) [20]. Contrary to the CP, TT allow different components of the decomposition to interact, resulting in a possibly more expressive decomposition. It could be that the system identification algorithm performs best for a TT model, which then needs to be converted into a reduced CP for stability analysis.

Another key aspect when dealing with large-scale power systems is the challenge to update the model. The power system, one of the most complex human build systems in the world, is constantly changing, new loads and generators connect and disconnect all the time, rendering any model that does not adapt outdated and thus inaccurate. We can identify the initial model via an alternating least-squares (ALS) procedure once an initial estimate for the state sequence is available. The first extension of this algorithm will be to find the unique reduced CP decomposition instead. Different optimization algorithms (ALS, subspace methods) for retrieving the tensor decompositions will be developed with a focus on accuracy and efficiency. Once an initial model has been identified and new data becomes available then this data can be used to update the model. This updating can be done by either a dual Kalman filtering approach or through a Block Coordinate Descent approach. Both approaches will allow real-time updates of the model.

SO5: Developing tools that are compatible with existing commercial software packages and new methodologies that are easily accessible.

To make the proposed tool more accessible to practitioners, academia, and end-user system operators like TSOs and DSOs, initial development, especially for extensions to the existing MTI toolbox [11], will be developed in MATLAB/Simulink which is a programming language and numeric computing environment well-established within the control systems and power systems community. We have a letter of intent (LOI) from MathWorks to ensure a smooth development process. For addressing the mentioned power system topics, an additional toolbox will be developed as a specially tailored extension, the TenSyGrid Toolbox. The development of this extension toolbox will consider further open access friendly languages like Python and Julia, prioritizing interfaces to existing widespread power systems software packages, like PSCAD, Digsilent PowerFactory, IPSA or PSS/E, and optimization solvers [21]. This compatibility criteria will be ensured by our industrial full partner eRoots, which is experienced in the usage and development of software for power system studies. We will ensure that the toolbox functionalities are in close relationship with our end-users

needs by keeping regular meetings with stakeholders like the Maltese DSO Enemalta, or the Korean Electric Power Corporation (KEPCO), which provided us with an LOI ensuring feedback and testing sessions for the toolbox functionalities. We also count with an LOI of the software provider IPSA (which provides Enemalta) for commercial compatibility and the real-time hardware provider Opal RT to improve the real-time feasibility of the toolbox.

1.5 Gender and diversity perspectives

No Gender biasing is expected from the proposed nature of work in this project, as the work in this project is independent of gender. Persons of all genders, identities and races without discrimination can perform proposed work and innovation in TenSyGrid. The results of this project can be implemented and exploited irrespective of gender. Nevertheless, gender dimension is considered by giving preference to the task force belonging to female and non-binary identities while recruiting new researchers within TenSyGrid. All TenSyGrid partners, i.e., Fraunhofer IWES, eRoots, HAW, UPC and UM have Gender Equality Plan (GEP) in place and it is available at their respective websites. The consortium is committed to implement gender equality in their respective organizations.

1.6 Open science approach

TenSyGrid aims to make the research carried in this project, data, and dissemination of results transparent and accessible to broad audience comprising of academia, industry, civil society and policy makers. This will also help disseminating the results and involve the power system community. Open science is a persistent part of WP6 on Knowledge community and WP7 on dissemination, communication, and exploitation.
The proposed key activities that contribute to open science in this project include:
Open access publications: The work program is structured in a way enabling publication of non-confidential research results in scientific journals as early as possible in fast peer-reviewed Open Research Europe or in high-impact journals following the green and gold access, such as, IEEE Access, IEEE Transactions on Power Systems, or IET, see Table 2.1 'Dissemination Activities'. Further, open access project publications could be accessed via online platforms such as OpenAIRE, Open research Europe, ResearchGate, Zenodo, Fraunhofer-Publica.
Open-source tools and software: TenSyGrid largely builds upon the existing MTI toolbox that is publicly available for Matlab. Therefore, TenSyGrid advances the existing toolbox as an open-source software tool in WP1 and WP 2 that allows other practitioners to utilize it also for other application domains. The current MTI toolbox is based on Matlab, but potential usage of other open-source software such as Python will be explored in WP4. eRoots uses and builds GridCal, an open-source Python-based package for the steady-state analysis of power systems. Dynamic functionalities are expected to be integrated soon there. Public repositories, such as GitHub, will be considered for toolbox development and uploading examples.
Sharing research data: TenSyGrid mostly deals with tools and software. Exchange of research data is based on FAIR principle promoted by EC. The relevant research data in machine-readable format and metadata will be stored at open access repositories and sharing networks e.g., GitHub.
Scientific collaboration: Facilitating knowledge exchange in WP6 to establish shared knowledge and via dissemination activities in WP7, collaboration and innovation in the scientific community bringing together diverse perspectives and expertise in the field is foreseen. It leads to more efficient and effective scientific processes and building of new projects to exploit results of TenSyGrid. The consortium seeks to engage with other projects during the project duration and will reach out if suitable projects have been identified. In particular, interactions with Consortium members other projects, such as CETP project WIND-DIGIPOWER of UPC and FLEXDIS, where eRoots and UPC are participating, is expected.
However, the topic of open science is complex to navigate from the commercial perspective. To comply with open science, adequate IPR protection measures are foreseen to safeguard business interests of partners. Before publishing, internal review process is followed to screen confidential information and protect IPR. This includes, but is not limited to deliverables, project reports, scientific or general articles, posters and presentations at workshops and conferences. There must not be a loss of competitiveness e.g., for the SME partner e-roots, or for the RTO IWES.

2. IMPACT

2.1 Expected outcomes and impacts

TenSyGrid will develop a novel toolbox for direct stability assessment for large-scale networks to achieve 100 % renewable energy. The TenSyGrid Toolbox as the main outcome will harness the capabilities of the

hybrid iMTI modeling class and its compact representation by decomposed tensors. The toolbox will impact the field of power system by providing new meaningful direct stability assessment methods to support real-time operation of very large power networks. The main target group of this project are TSOs & DSOs, enabling them to integrate more renewables energies in their networks. This ambitious outcome is achieved by models that find a compromise in modeling accuracy. The goal is to not necessarily finding the most optimal solution to maintain a stable network, but a sufficiently good sub-optimal solution that can be computed faster while being accurately enough. The TenSyGrid project is the next step in the development to enhance the existing MTI Toolbox by extending its capabilities and tailoring it to the need of power systems. The TenSyGrid Toolbox as the outcome of this pathway is a milestone towards developing commercial solutions at a later stage.

TenSyGrid aligns with the ambitions of the revised European Strategic Energy Technology (SET) Plan. The SET plan aims to accelerate the development of innovative and flexible solutions that optimize the existing energy grid. By providing cutting-edge assessment methods, this project empowers decision-makers to make informed choices that enhance the stability and security of the grid. This optimization directly impacts the market by enabling a more efficient response to demand in a systematic manner. Ultimately, this benefits consumers while also addressing urgent societal needs and climate protection. TenSyGrid is in line with the scope of the European Green Deal and the CETP- instrumental to the SET Plan. The SET Plan's priorities include making the EU a leader in renewables (Priority 1) and delivering a smart, consumer-centric energy system (Priority 2). While TenSyGrid supports the vision to increase the share of overall EU renewable electricity production integrated into the grid to at least 65% by 2030, it aims at networks across the globe to achieve 100 % renewable energies.

The outcomes stemming from the successful implementation of the SOs described in the previous section will lead to several impacts across all stakeholders of TenSyGrid. The foreseen immediate stakeholders are end-users like TSOs & DSO system operators, engineering offices and consultancies, manufacturers, software providers, practitioners in general as well as all academics in the related fields. With regard to projects specific objectives these impacts are:

- SO1: Its fulfillment will result in a **model library of hybrid iMTI models** that capture the relevant nonlinearities of different power system components. The quality of the library will be ensured by our partner, the Polytechnic University of Catalonia (UPC), who will leverage the information and contacts available across our transnational consortium. Such a model library will have a significant scientific impact across all stakeholders (commercial and academic) by enabling more computationally efficient analytic studies and (indirectly) simulations than general nonlinear EMT studies and more accurate than linear approximations.
- SO2: Its successful implementation will result in a **package of novel real-time methods for power system direct-stability assessment**. These methods will have a direct technological and economic impact on system operators by, for example, increasing the limit of instantaneous VRE penetration by providing a better stability margins estimation than the benchmark or aiding in critical scenario selection. Moreover, the accuracy of the methods can be adapted by tuning the rank of the tensor decomposition approximation allowing for better computational performance improving, e.g., offline assessments like grid planning, or like day-ahead transient stability studies, especially for converter interactions. This will add a significant scientific and technological effect that will reach not only system operators, but power users like consultancies, manufacturers, and even individual users like practitioners and academics. Furthermore, the transnationality of the consortium will be a source of different grid topologies, energy matrices, and regulatory requirements, to enhance the developed methods applicability. For example, the Jeju Island in South Korea is expected to be a test field candidate through our cooperation with our industrial advisor KEPCO (See section 2.3).
- SO3: We expect the development of an **interpretable system identification method based on decomposed hybrid iMTI models**. The interpretability of iMTI models will have scientific, economic, and technological impacts for system operators and manufacturers, improving the assessment of the impact of OEM assets for grid integration, e.g., for converter interaction studies. Similar to the SO1 outcome, access to asset information from across our transnational consortium will improve the reliability of the developed method.
- SO4: It complements the outcome of SO1 by making it easily scalable and updatable. This will have a direct scientific and technological impact for system operators, enabling them to take more **informed decisions in real time based on truly up-to-date models of the grid**. Beyond system operators, this outcome will have a scientific impact on all stakeholders, allowing for large analytic studies with adaptable accuracy to performance ratio.

- SO5: We expect to extend the power system stability analysis capabilities of existing commercial software like, for example, IPSA. This will naturally have a direct economic impact on the value proposition of software providers. However, due to the widespread nature of these software technologies, we expect to have technological and economic impact across all stakeholders, for example, by providing **better planning and operation tools through the enhanced commercial software offer.** Moreover, the widespread nature of these tools will also contribute to overcoming one of the main barriers for the impact of this project, reaching the end-users. In this context, the cooperation with our industrial associate partner KEPCO from South Korea, will enable us to consider the software needs in a transnational scale, complementing the European power system perspective of the other partners.

2.2 Transnational added value

The transnational collaboration within the TenSyGrid project significantly enhances the value and impact of its research activities, setting a precedent for innovative approaches to managing the complexities of power systems with high penetration of power electronic-interfaced power sources (HPoPEIPS). This aspect of the project is a critical component, reflecting the collective expertise and resources of the participating entities from Germany (IWES, HAW), Spain (UPC, eRoots), and Malta (UM), each bringing unique insights and capabilities (refer section 3.3.) to the table. Besides, we have advisory board members spread across three continents from countries Netherland (TU Delft), Malta (Enemalta), South Korea (KEPCO), France (RTE) and Germany (MathWorks, OPAL-RT). The added value of such collaboration is multifold, encompassing the broad spectrum of technical, scientific, and cultural perspectives that enrich the project's outcomes. This fosters a richer exchange of ideas and best practices, encouraging a cross-pollination of knowledge that is vital for driving innovation. One of the primary benefits of this transnational partnership is the pooling of diverse expertise and knowledge bases, which facilitates a more comprehensive and nuanced approach to the challenges at hand. By leveraging the specialized skills and experiences of its partners, TenSyGrid is positioned to develop more robust and innovative solutions that are not confined by the limitations of a single national context. Power grids may have varying legislation behind them depending on the national grid operator. Collaborating with partners from different countries, allows for the generation of more generalized solutions for the multifaceted power system analysis in a converter-dominated energy landscape, ensuring the development of methodologies and tools that are adaptable and effective across different regulatory, technical and environmental scenarios.

2.3 Dissemination and communication

We perceive a general awareness among the relevant stakeholders about the need for more accurate tools that can easily scale to large networks with HPoPEIPS. Furthermore, we note that there is a strong focus on computational improvements of time-domain simulation instead of looking for other approaches. That is also partly owed to the fact that nonlinear system theory is not necessarily straightforward and requires different expertise. This applies to TenSyGrid, where knowledge about multilinear models, tensor algebra, and general nonlinear system theory is required besides that of power system engineering. Therefore, TenSyGrid will, firstly, share the developments and new findings with the research community, secondly, interact with the power system community and relevant stakeholder to address the right problems, and thirdly, involve the end-users to share the outcomes and to build trust into the methods to maximize the impact.

TenSyGrid will benefit by the involvement of the Knowledge Community (WP6) and the implementation of an advisory board. The project has gathered significant interest from large enterprises and academia, reflecting the potential of TenSyGrid. The **members of the advisory board** are:

- **Enemalta plc** is the leading energy services provider in the Maltese Islands, entrusted with the distribution of electricity, and the development of the national electricity distribution network.
- **KEPCO**, South Korea’s largest energy provider, is responsible for electricity generation, transmission, and distribution.
- **OPAL-RT**, a market leader in real-time simulation of electrical systems, brings expertise in power systems and electronics real-time computation.
- **MathWorks** is a software company that develops mathematical computing software, most known for MATLAB and Simulink, used for modeling within TenSyGrid.
- **Réseau de Transport d'Électricité (RTE)** is the French transmission system operator and is continuously looking for new approaches.
- **TNEI (IPSA Software Package)** – It is a power system analysis software tool owned by the specialist

energy consultancy, TNEI. L

- **Delft Tensor AI Lab at the Delft University of Technology** represented in project by the co-director Kim Batselier. The focus of his research group lies on the development of new theory and algorithms for efficient nonlinear system identification using tensor decompositions.

The letter of intent from the respective advisory board members can be provided to the funding agency, if asked for after successful evaluation of the project.

These companies and TU Delft form the project's advisory board, which will provide important input on identifying critical scenarios, providing feedback to the developments, and helping in the exploitation of the results. Besides their importance on the project development, they will help in disseminating the project results to increase the impact, for example, as RTE could report the results to ENTSO-E, the European Network of Transmission System Operators. In addition to the advisory board, the Knowledge Community will provide useful feedback and will serve as a good platform to disseminate the project results to maximize the impact. Combined with KEPCO, TenSyGrid impact can extend to different areas of the world.

TenSyGrid has several dissemination measures as part of WP7 that are summarized in Table *1***Fehler! Verweisquelle konnte nicht gefunden werden.**, which will be maximized by the active involvement of each partner. These activities aim at different stakeholder groups. During the project and transitioning from a TRL of 3 to 5, the activities will be focused on reaching relevant researchers by conference presentations, open access journals, sharing other material via relevant platforms, e.g., a GitLab repository, to disseminate the results but also to get valuable input. This goes along with the activities of the advisory board with workshop, telecons, as well as the engagement of the knowledge community. Besides a project webpage, relevant communication channels and stakeholders will be addressed by LinkedIn. For the engagement of early-career researchers as well as potential end-users, a hackathon is planned. These activities will be updated with the progress of the project.
As mentioned in the SO5, the goal of TenSyGrid is to provide useful tools that address the right problems to enable 100 % renewables and that are accessible by daily practitioners. For this reason, TenSyGrid wants to demonstrate the tool capabilities in a relevant commercial software environment at the end of project in WP5. A final public workshop will present these results to the important stakeholders where the involvement of the advisory board will be important. The toolbox will be under an open-access license, such as the MIT license. For the interface to commercial or third-party software other more restrictive licenses might be required. The licensing and open science practice will be addressed in more detail in the full proposal.

Table 1: Summary of dissemination measures

Channel	**Timing (Month)**	**Target group**
Press Release	M1-M36	All stakeholders, public (min. 4 releases)
Webpage	M4	Industry, Academia, Civil society and Policy makers
LinkedIn and X	M1-M36	Social media audience targeting industry and researchers
Peer-reviewed publications (open access)	M9 – M36	Scientific community and R&D from TSOs and DSOs (min. 1 publication/ academic partner) for example, IEEE Transactions on Power Systems, IEEE Access, Automatica
Presentation at conferences (oral/poster)	M6 – M36	Scientific community and R&D from industry incl. TSOs & DSOs (2 conferences /partner), for example, PSSC 2026, PowerTech 2027, ECC 2026
Open source & open access data (git, arXiV)	M6 – M36	Scientific community
Project summary	M36	Scientific community, industry, and decision makers
Hackathon-type activity	M17	Early career researchers in the research field of power system stability analysis
Knowledge Community	M1 – M36	Participation of every partner in the working groups
Kick-off and annual meetings with the advisory board	M1 – M36	Advisory board members (academia and industry), feedback and dissemination of results
Scientific workshop on MTI modelling	M25	Workshop on multilinear modeling and practical stability analysis for power systems
Stakeholder seminar	M36	Presenting final results to relevant stakeholders and advisory board members, demonstrating the tools and

		the implications for real-time assessment

2.4 Exploitation

The TenSyGrid project sets ambitious targets for the exploitation of its outcomes, aiming to significantly impact both the scientific community and the energy sector. With its innovative approach to power system stability analysis through the development of a new modeling framework and real-time assessment tools, TenSyGrid expects to deliver solutions that are of substantial significance beyond the project timeline. The project emphasizes the compatibility of its tools with existing commercial software packages (such as PowerFactory, PSCAD, IPSA, etc.), ensuring that the innovations developed can be readily adopted by a large pool of industrial users utilized by industry practitioners, particularly transmission and distribution system operators (TSOs & DSOs). The project aims to deliver a packaged tool to be further implemented on grid operators' control rooms and employed on a daily basis to operate modern power networks. Additionally, partnerships with software developers can be derived from this project given the fact that the proposed multilinear approach supposes a change of paradigm in the analysis of power systems.

The exploitation strategy is designed to ensure that the project's outcomes are not only academically relevant but also practically applicable, facilitating a smoother transition towards grids with higher penetrations of renewable energy sources. By addressing the current limitations of existing models and methodologies, TenSyGrid aims to provide system operators with more reliable and interpretable tools for real-time operation and planning. The project's focus on developing scalable, easily updatable models and tools shows its commitment to a long-term exploitation strategy, promising a lasting impact on how power systems are analyzed and managed in the context of the energy transition. All in all, the proposed approach has the potential to become the de facto methodology for analyzing power grids.

Table 2: Key exploitable results (KER) and desired outcomes

Exploitable results			**Exploitation strategy**		**Desired outcome**
No	**Result**	**WP**	**Target group**	**Exploitation action**	
KER1	Hybrid Implicit Multilinear Modelling Class	1	R&D institutes, Software providers	Presenting the capabilities of hybrid implicit multilinear models and suitable tensor decomposition	Greater visibility of multilinear models and advantages for certain fields, e.g., power system, or heating sector
KER2	Alpha version of the MTI Toolbox	1, 2	R&D institutes	The alpha version of the MTI Toolbox is distributed to a small number of researchers for testing and feedback	Identification of problems and bugs, potential users, and improving the visibility of the MTI Toolbox
KER3	MTI Toolbox final version	1, 2	R&D institutes, Software providers	The final MTI Toolbox is distributed to different stakeholders, such as university researchers and software developers.	New users of the MTI Toolbox, in different fields of applications other than power systems
KER4	Multilinear Power System Models	3	R&D institutes Power System Engineers	Showcasing and promoting multilinear models of relevant power system components to efficiently address their nonlinearities and saturations	Capabilities Advantages of multilinear modelling are leveraged by relevant power systems related users and stakeholders
KER5	Alpha version of the TenSyGrid Toolbox	4	R&D institutes	The alpha version of the TenSyGrid Toolbox is distributed to a small number of researchers for testing and feedback	Issue detection, and fostering the visibility of the MTI TenSyGrid Toolbox
KER6	Power system	4	TSOs &	Previous to a working	Feedback, industrial

	analysis based on multilinear models		DSOs, Engineering offices, Commercial software providers	version of the Toolbox, showing the results of power system analysis using multilinear modelling and stability assessment beyond linear models to support decision makers	needs, and interest in obtaining these capabilities of TenSyGrid
KER7	Benchmarking studies of the TenSyGrid Toolbox	4,5	TSOs & DSOs, Engineering offices, R&D institutes	A full showcase of the TenSyGrid Toolbox from: a) a fundamental and academic perspective, and b) a comparison with common commercial tools, leveraging the integration of the Toolbox into commercial solutions	Greater visibility and demonstration of the capabilities of the TenSyGrid Toolbox
KER8	Final version TenSyGrid Toolbox	4	TSOs & DSOs, Engineering offices	A demo case study and the final version of the TenSyGrid Toolbox is distributed to a selected number of important end-users and stakeholders	Feedback to the toolbox developers and suggestions for extensions, interest in usage and follow-up activities for commercialization

The TenSyGrid toolbox developed in this project serves as a valuable tool to assess grid stability. It is meant to be employed in all kinds of networks, yet its advantages can be more easily exploited in modern networks based on power electronics, such as those including renewable energies of varied nature, storage solutions, among others. The technology developed through this project holds significant relevance for the market. For instance, the study conducted on the Malta model can be replicated in other geographical locations, making it applicable and adaptable to different regions e.g. in Jeju islands of Korea. Also, eRoots will employ the toolbox in its own studies and promote it to its clients, who will have been able to assess beforehand the advantages of relying on it. Overall, the toolbox will showcase a high level of reproducibility, replicability, and interoperability.

Being built on open-source principles, the Toolbox enables further contribution from other developers and researchers by possibly adding new and/or enhancing features. In this direction:

- The toolbox can be used as a basis for collaborative research projects with academic institutions or other research organizations. This can help advance the state-of-the-art in the field and lead to new discoveries and innovations.
- Building a community of engineers around the toolbox through forums, user groups, and online communities can help foster collaboration, knowledge sharing, and user engagement. This can lead to valuable feedback, contributions, and advocacy for the toolbox.
- Training courses or educational materials can be offered to help users learn how to effectively use the toolbox and its potential in analyzing modern power systems. This can include online tutorials, webinars and workshops.
- The toolbox can be integrated into numerous power system analysis software packages to enhance their capabilities in assessing power networks.
- The toolbox will only contain open component models. This exercise will bring transparency, facilitate the model debugging process, and ease the information transfer process. Model formats such as .cgmes, .raw, .dgs, .GridCal will be supported.
- The toolbox could be partially distributed under an open license. For example, licenses such as GPL and AGPL would allow users to use, modify and distribute the software for free if they comply with the license terms. Users who do not want to comply with the terms and avoid sharing the modifications with the community could acquire a commercial license, which could bring economic viability to the Toolbox. The next section elaborates on it.
- While the models and main algorithms can be open source, the partners of the project could also

commercialize the toolbox by building a rich graphical user interface to improve the user experience, hosting the application in the cloud, or adding premium functionalities that boost the performance.

The project forms an advisory board (see section 2.3) with eminent industry members. This board will help steer the technology towards demonstration and eventually commercial exploitation of project results beyond the end of the project, potentially in the next 5 years. Additionally, they assist in examining and providing guidance on research activities throughout the project and dissemination of project results that would attract investment.

2.5 IPR management

In line with the collaborative nature of the TenSyGrid project, the consortium is committed to develop a comprehensive strategy for Intellectual Property Rights (IPR) management This strategy encompasses clear guidelines on access to background knowledge and data, as well as procedures for the protection of innovations through suitable forms, such as patents, licenses, trade secrets as applicable for the project findings. Data transfer, storage and processing are important for TenSyGrid. The management of data is guided by Findable, Accessible, Interoperable and Reusable FAIR framework to ensure smooth data exchange between entities. The management of data, accessibility and platform for open access will be discussed among partners to form the data management strategy. The consortium intends to recognize important project results and check exploitation measures at plenary meetings, ensuring that all partners are aligned on the exploitation and the protection of intellectual property. The IPR & data management strategy is presented as part of deliverable 7.2 i.e. plan for communication and exploitation of the results. The status of expected results is updated and reported annually in the project reports.

At the end of project, it is expected to reach minimum TRL 5 which implies further development of toolbox for industrial requirements before the commercialization of the product. Nevertheless, protection measures for future commercialization is considered in TenSyGrid. The consortium aims to open the code of the developed tools throughout the project to facilitate the exchange of information between the partners and the outside contributors/users. The protection of information can be related to additional functionalities or improvements in the performance. For this scenario, the project strategy is to adopt dual licensing approach i.e. to adopt two licenses:

- Open license: the software is released under an open-source license such as the General Public MTI License. The community can use the code, modify it, and redistribute it, while maintaining the same open-source license. This way it is ensured that the modifications remain open and available to the community, thus advancing the original code.

- Commercial license: it allows modifying the code and integrating it into the platform of the client without the need to disclose the source code nor share the modifications. On top of that, the commercial license of Toolbox could include additional features, such as a parallelization engine, a responsive graphical user interface, automatization scripts, among others.

From a business standpoint, the goal is to offer open models and algorithms to everyone (free of charge) plus commercial functionalities for specific companies (premium). Thus, it is believed that this freemium model would help at attracting a large user base and securing revenue from serious users.

At successful evaluation of TenSyGrid proposal, a Consortium Agreement outlines the process for managing the knowledge produced by TenSyGrid. If there is joint ownership of the intellectual property (IP), a separate contract will be drafted and signed by the co-owners to define their rights and responsibilities and address the terms of commercial licensing.

2.6 Contribution to the clean energy transition

Clean energy transition calls for the development and deployment of advanced grid management tools and technologies for resilient energy systems. Integrating VRE into the existing power grid poses challenges due to their fluctuating nature. Efficient assessment of grid stability is crucial to ensure smooth and reliable integration to achieve global imperative that aims to shift away from fossil fuels and embrace renewable energy sources for overarching climate goal. TenSyGrid aims for improved stability assessment to evaluate real-time power grid operation, though offline solutions are not discarded, which also extend to future grid development plan scenarios. The end-user will be able to define more ambitious pathways towards the 100% renewable energy system, i.e., a higher penetration of VRE, given the stability margins for different grid development plan scenarios. The international consortium ensures that the usability of the tool is not grid specific but works for every grid that is transitioning towards 100% renewable energies.

TenSyGrid benefits from the involvement of the SME eRoots and the grid operators KEPCO, RTE, Enemalta,

to ensure usability for the end-users i.e., grid operators, policy makers, or local energy system designer/operator. It also benefits from the involvement of MathWorks and OPAL-RT for ensuring highly efficient programming. Both are bringing in the competencies of large enterprises on real-time simulation. And it benefits from the involved interdisciplinary researchers and project advisors as TU Delft for excellence in tensor calculus, usage of tensors in control engineering, and power system simulation.

3. Quality and efficiency of the implementation

3.1 Work plan outline

3.1.1 Project structure

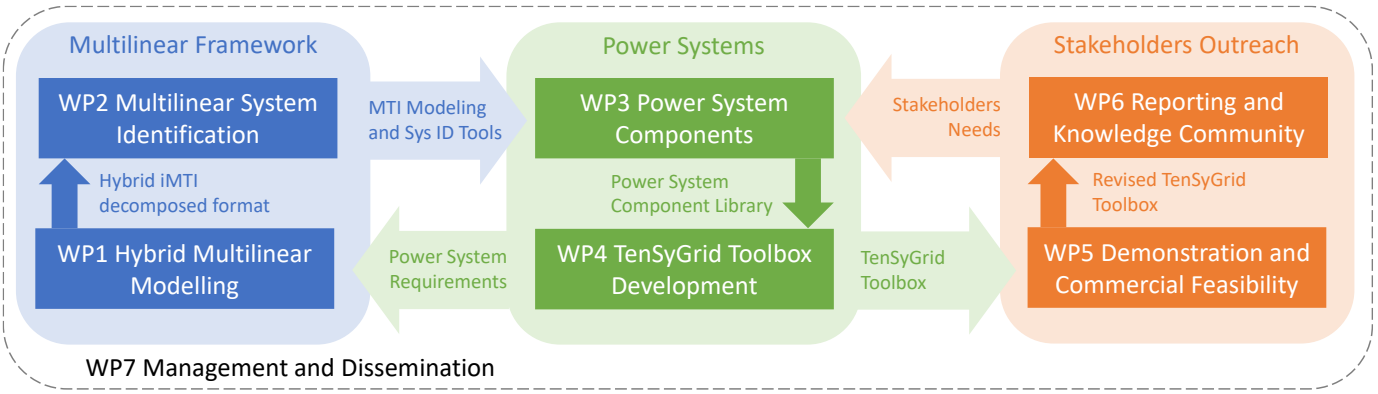


Figure 6: Work packages interrelation chart.

Table 3: Summary of all staff effort

Partner	WP1	WP2	WP3	WP4	WP5	WP6	WP7	Total person months per participant
1 IWES	12	7	2	6	2	1.5	2.5	33
2 UPC	4	2	6	12	4	1	1	30
3 HAW	10	17	2	4	1	1	1	36
4 eRoots	4	2	14	8	6	1	1	36
5 UM	1	3	5	7	18	1	1	36
Total person months	31	31	29	37	31	5,5	6,5	171

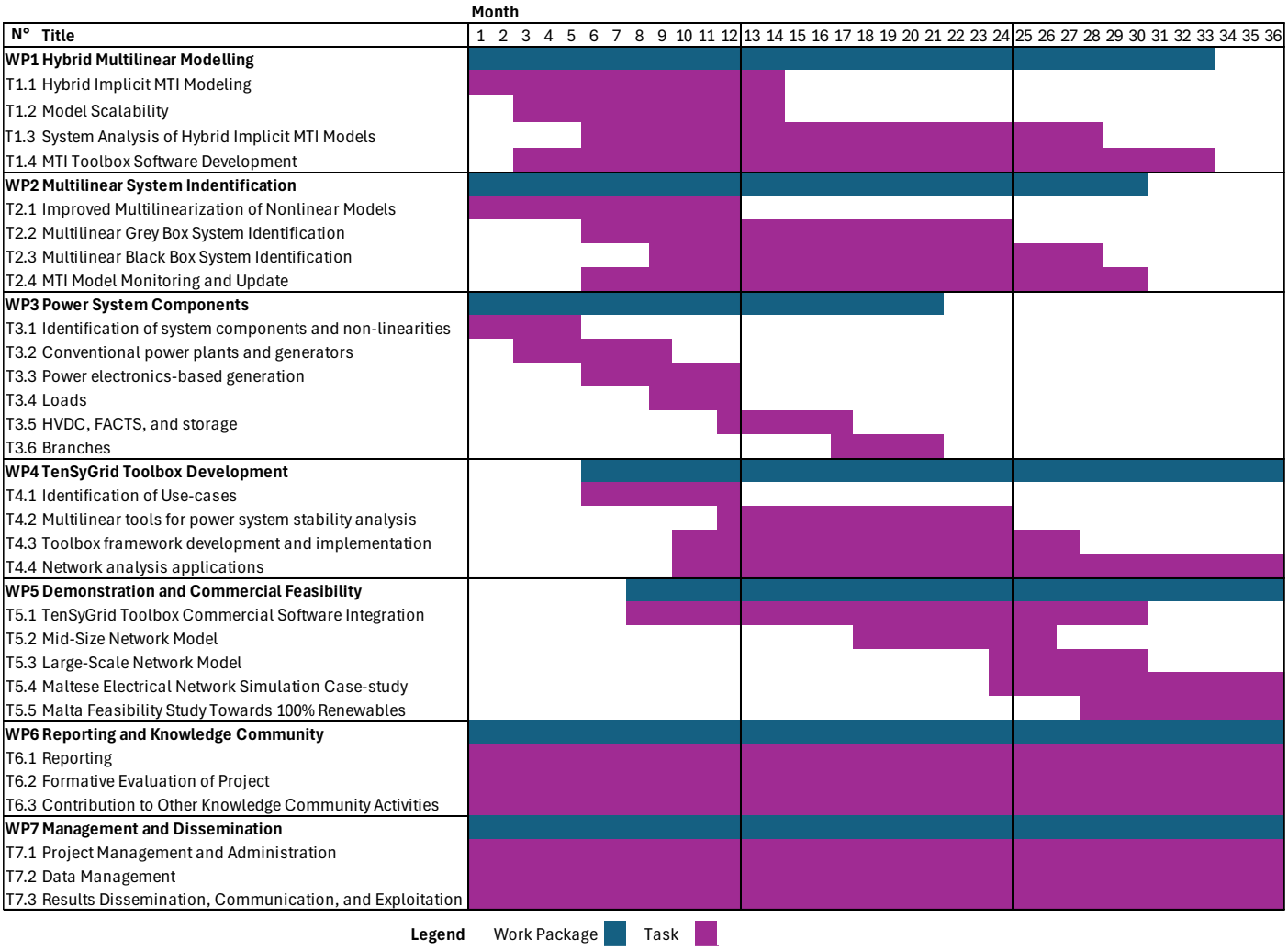


Figure 7 Work Plan Gantt Chart

3.1.3 Work packages

WP1 Hybrid Multilinear Modeling (Lead: IWES)

Work package number	1	**Lead beneficiary**			IWES	
Work package title	**Hybrid Multilinear Modeling**					
Participant number	1	2	3	4	5		
Short name of participant	IWES	UPC	HAW	eRoots	UM		
Person months per participant:	12	4	10	4	1		
Start month	1		**End month**	33			

Objectives

WP1 focuses on extending the existing multilinear modeling theory, prioritizing the development of hybrid implicit models. Continuous-valued explicit MTI models were introduced as expansion of linear state space models a decade ago and have shown advantages in some application areas like HVAC systems since then. Implicit MTI models are a closed subclass of nonlinear differential algebraic equations (DAEs) allowing multiplications of state derivatives with states, inputs, and outputs. Moreover, hybrid iMTI models include discrete valued signals like binary switching. The extended capabilities of the hybrid iMTI class are essential for capturing the most predominant nonlinear dynamics of power systems like saturations for stability assessment. The goal is to extend the MTI toolbox with the necessary theoretical framework for the composition, decomposition, efficient storage, and simulation of hybrid iMTI models. The main objectives are:

- Extend the multilinear modeling class to hybrid implicit multilinear models.
- Develop tensor representations for multilinear dynamical models which are composable and

decomposable.

- Find minimal representations, i.e., suitable reduced rank approximations, to ensure scalability of MTI models for large systems (SO4).
- System analysis of hybrid implicit multilinear models to enable stability analysis of power systems

Description of work

T1.1 Hybrid Implicit MTI Modeling (**IWES**, HAW, UPC, eRoots, UM) M1-M14: The existing multilinear modelling class is extended to find multilinear representation of a set of general nonlinear equations (SO1) e.g., trigonometric functions, piecewise linear functions, Boolean functions, etc., based on the requirements from WP3. For these, the index of the hybrid implicit multilinear model, which are DAEs, is determined to choose appropriate numerical solvers.

Distribution of work: IWES leads the development of the multilinear models in sparse decomposed forms with feedback from the HAW for tensor decomposition competency. Input comes from eRoots and the UPC out of WP2 for the required dynamics from power system examples with support from UM.

T1.2 Model Scalability (HAW, IWES, UPC, eRoots, UM**)** M3-M14: Based on the outcome T1.1, composable model representations in sparse decomposed minimal forms are developed and theoretically described. This includes looking for suitable tensor decomposition techniques that ensure updateability and scalability (SO4) of the model. Rank approximations are tested for a number of examples to find a minimal representation of the model with sufficient accuracy.

Distribution of work: HAW leads the development for composable model in close collaborations with IWES. eRoots and UPC provide feedback to the accuracy of models for rank approximations with support of UM.

T1.3 System Analysis of Hybrid Implicit MTI Models (**IWES**, HAW, UPC, eRoots, UM) M6-M28: Methods for system analysis of MTI models as decomposed tensors are developed similarly to the coefficient analysis of matrix decomposition in the linear case. Linearization of hybrid implicit models for classical eigenvalues analysis will be created, and other concepts, e.g., for approximations of region of attractions of fixed points by cell mapping, or determining stability margins, are explored to provide relevant information about the system (SO2).

Distribution of work: IWES leads the activity in close collaboration with HAW and input from UPC, eRoots and UM on important for stability checks for power systems. Therefore, this tasks closely relates to T4.1 and T4.2 to provide the capabilities for the TenSyGrid Toolbox.

T1.4 MTI Toolbox Software Development (HAW, IWES, UPC, eRoots, UM**)** M3-M33: All developments from the previous tasks and WP2 are added to the existing MTI Toolbox to extend its capabilities. From alpha to a polished version, the MTI Toolbox will be incrementally updated and improved during throughout the project with dedicated sprints to achieve a readiness for the TenSyGrid Toolbox.

Distribution of work: HAW leads the MTI Toolbox development in close collaborations with IWES. UPC, eRoots, and UM provide feedback on capabilities and usability of the toolbox and support in the documentation.

Deliverables (PU: Public, CO: Confidential)

D1.1 Hybrid Implicit Multilinear Model Class and Minimal Format (IWES, PU, M12): This software and document establishes the hybrid multilinear modelling class including tensor decomposition techniques.

D1.2 Alpha Version of the Updated Multilinear MTI Toolbox (IWES, CO, M20): This software and documentation enables the system analysis of hybrid multilinear models aimed for power system analysis at a later stage.

D1.3 Final Version of the Updated Multilinear MTI Toolbox (HAW, PU, M33): The MTI Toolbox is updated and a documentation of the new capabilities with some examples is provided.

Milestones

MS1.1 Hybrid iMTI Decomposed Form (IWES, M14): Review the functionality with mid-size examples of WP3 uploaded into a public repository.
MS1.2 Analytic Stability Assessment in MTI Form (IWES, M24): Review the functionality with mid-size examples of WP4 uploaded into a public repository.

WP2 Multilinear System Identification (Lead: HAW)

Work package number	2	**Lead beneficiary**			HAW	
Work package title	Multilinear System Identification					
Participant number	1	2	3	4	5	
Short name of participant	IWES	UPC	HAW	eRoots	UM	
Person months per participant:	7	2	17	2	3	
Start month	1		**End month**	30		

Objectives

In WP2, the focus is on interfacing the MTI toolbox with the real-world power system, making the toolbox accessible to end-user applications. Typically, power electronics models are shared in a black-box manner with operators, enabling a simulation-oriented study but complicating an analytical approach. This motivates the goal to provide the toolbox with system identification capabilities that can generate decomposed hybrid iMTI models directly from measurement data of the real system or simulations of proprietary OEM black-box models addressing SO3. Depending on system information availability, multilinearisation or grey/black-box identification can be pursued. The main objectives are:

- Develop methods for the multilinearization of general nonlinear models.
- Develop system identification methods for hybrid implicit multilinear models.
- Develop detection and update methods for parameter and structural model changes.

Description of work

T2.1 Improved Multilinearization of Nonlinear Models (**HAW**, IWES, UPC, eRoots, UM) M1- M12: Improve the existing multilinearization capabilities of the MTI toolbox to support large-scale sparse hybrid nonlinear systems. Moreover, the multilinearization into firstly semi-explicit and later implicit hybrid MTI models should also be supported.

Distribution of work: HAW leads the multilinearization method development in close collaboration with IWES. UPC, eRoots, and UM provide feedback base on use cases on WP3 and WP5.

T2.2 Multilinear Grey Box System Identification (**HAW**, IWES, UPC, eRoots, UM) M6- M24: Develop new system identification methods aim towards systems with a known or assumed underlying multilinear structure. The focus is the parameter identification of such systems for explicit, semi-explicit, and implicit formulations of hybrid MTI systems.

Distribution of work: HAW leads the system identification methods development in close collaboration with IWES. UPC, eRoots, and UM provide feedback base on use cases on WP3 and WP5.

T2.3 Multilinear Black Box System Identification (**HAW**, IWES, UPC, eRoots, UM) M9- M28: Develop new system identification methods aim towards systems with an unknown underlying structure, e.g., OEM black-box models, addressing SO3. The focus is finding the most suitable underlying multilinear structure that better captures the system dynamics with a tuneable complexity and footprint trade-off, following SO1. Furthermore, support is expected for explicit, semi-explicit, and implicit formulations of hybrid MTI systems.

Distribution of work: HAW leads the system identification methods development in close collaboration with IWES. UPC, eRoots, and UM provide feedback base on use cases on WP3 and WP5.

T2.4 MTI Model Monitoring and Update (**HAW**, IWES, UPC, eRoots, UM) M6- M30: Enhance current MTI methods for fault detection to develop new methods for the monitoring of MTI models that enables the detection of parameter changes or structural changes. The focus is on triggering the update of the MTI model by efficient tensor operations, following SO4.

Distribution of work: HAW leads the monitoring and update methods development in close collaboration with IWES. UPC, eRoots, and UM provide feedback base on use cases on WP3 and WP5.

Deliverables (PU: Public, CO: Confidential)

D2.1 Improved Multilinearization Algorithm (HAW, PU, M12): Provides the software and documentation for the new multilinearization algorithm of the MTI Toolbox.

D2.2 Multilinear Grey Box System Identification Algorithm (HAW, PU, M24): Provides the software

and documentation for the new multilinear grey box system identification algorithm of the MTI Toolbox.

D2.3 Multilinear Black Box System Identification Algorithm (HAW, PU, M28): Provides the software and documentation for the new multilinear black box system identification algorithm of the MTI Toolbox.

D2.4 MTI Monitoring and Update Algorithm (HAW, PU, M30): Provides the software and documentation for the new monitoring and update algorithms of the MTI Toolbox.

All as contribution towards D1.3

Milestones

MS2.1 System Identification in Decomposed MTI Form (HAW, M21): Review the functionality with mid-size OEM examples of WP5 uploaded into a public repository.

WP3 Power System Components (Lead: eRoots)

Work package number	3	**Lead beneficiary**				eRoots	
Work package title	Power System Components						
Participant number	1	2	3	4	5		
Short name of participant	IWES	UPC	HAW	eRoots	UM		
Person months per participant:	2	6	2	14	5		
Start month	1	**End month**		21			

Objectives

WP3's goal is to compose a library of standardized models of the most relevant power system components (e.g., synchronous generators, wind turbines, power converter, power lines) in decomposed hybrid iMTI format for the MTI toolbox. The structure and dynamics of some of these models can be used as the foundation for grey box modeling parameter identification of WP2. The idea is to identify the critical nonlinear dynamics that need to be captured to accurately assess the power system stability. Some of these nonlinearities will be natively captured by the hybrid iMTI format, like saturations; however, some will require further development to achieve equivalent behavior, e.g., through change of variables to systems with augmented states like for the Park transform.

- Select a comprehensive set of power system components and develop their underlying equations with focus on non-linearities and saturations.
- Develop multilinear models for relevant power system components.
- Validate the developed models and ensure that they capture all relevant non-linearities.

Description of work

T3.1 Identification of System Components and Non-linearities (eRoots, UPC, IWES, HAW) M1-M5: A first screening of the main equations related to power grids will be performed. This includes the steady-state Kirchhoff power balance equations, but also the particularities of dynamic models. Among these dynamic models, transformations implemented in power converters such as the Park and Clarke transformation have to be considered, and also the saturation of devices (current saturation in transformers, capability curves in generators, etc.). The goal is to envision the main complications the equations can pose to the transformation of the original models into multilinear models.

Distribution of work: eRoots and UPC indicate the most relevant equations in power system components and identify the most critical non-linearities and saturations, together with the help of IWES and HAW.

T3.2 Conventional Power Plants and Generators (**eRoots**, UPC, UM, IWES) M3-M9: The task will develop the multilinear models of the specific elements employed in traditional power plants. A simplified conventional (thermal/hydro) power plant will be modelled. Focus will be placed on the dynamics of the synchronous generator, including the exciter, governor, and turbine models. All the developed models will be validated with realistic nonlinear models.

Distribution of work: eRoots takes charge of developing the models for conventional generators and power plants. IWES supervises their expression in multilinear fashion, while UPC and UM work on the validation.

T3.3 Power Electronics-based Generation (eRoots, UPC, UM, HAW) M6-M12: The task will cover

the expression of power electronics devices' behavior used to interconnect renewable energy sources into the grid in a multilinear manner. The main control blocks of converters, such as the inner current control loop, the voltage control loop, the phase-locked loop, the Park and Clarke transformations, among others, will be transformed from a non-linear form with potential saturations into a multilinear representation. The performance of the multilinear model will be tested and validated by running various EMT simulations in conventional software that handles non-linear systems from a dynamic standpoint.

Distribution of work: eRoots and UPC detail all transformation and control blocks, HAW will aid on the conversion to multilinear models, and UM will work on the validation in a standard power grid.

T3.4 Loads (**UPC**, eRoots, UM, HAW) M9-M12: The task will address the multilinear modelling of several types of common loads and aggregations of loads. The models will be validated with realistic data obtained from several domestic and industrial loads. Some examples will include aggregated loads over large areas and more detailed loads in large buildings, industry or electrical vehicle charging stations.

Distribution of work: UPC and eRoots will model the loads considering their dynamics, UM provides realistic data to validate the models, and IWES helps at the transformation of non-linear to multilinear models.

T3.5 HVDC, FACTS, and Storage (**eRoots**, UPC, UM, IWES) M12-M17: The task will focus on energy storage solutions Flexible AC Transmission Systems (FACTS) and High Voltage Direct Current (HVDC) technologies. Several multilinear models will be developed and validated, including large power converters for HVDC and FACTS, electrolyzers, and batteries.

Distribution of work: eRoots and UPC review the conventional models for storage solutions, HVDC and FACTS, UM works on their validation, and IWES collaborates with eRoots on the multilinear model expressions.

T3.6 Branches (**eRoots**, UPC, UM, HAW) M17-M21: The task will develop and validate multilinear models for branches, such as power transformers, overhead lines, and underground and submarine cables (including AC and DC options). Line and cable models will consider frequency dependent behavior. Transformer models will consider their inherent non-linearities, including the effects of taps on the resulting admittance matrix and potential saturations.

Distribution of work: eRoots proposes the set of models to analyze, UPC develops multilinear models for transformers, underground and submarine cables, and overhead lines, UM validates multilinear models for underground and submarine cables, and IWES helps UPC and UM on the model validation.

Deliverables (PU: Public, CO: Confidential)

D3.1 Report on Identified Non-linearities and Saturations (eRoots, PU, M5): Listing of the equations and potential complications when transforming them to multilinear expressions.

D3.2 Generation Multilinear Models and Validation (eRoots, PU, M12): Creation and validation of multilinear models for generation components, both traditional and power electronics based.

D3.3 Loads, Branches, Storage, FACTS and HVDC Multilinear Models and Validation (eRoots, PU, M21): Creation and validation of multilinear models for non-generation devices such as loads, branches, storage, FACTS and HVDC.

Milestones

MS3.1 Power System Component Models Completed (eRoots, M21): Models validated comparing to complete non-linear classical models and uploaded into a public repository.

WP4 TenSyGrid Toolbox Development (Lead: UPC)

Work package number	4	**Lead beneficiary**			UPC	
Work package title	TenSyGrid Toolbox Development					
Participant number	1	2	3	4	5	
Short name of participant	IWES	UPC	HAW	eRoots	UM	
Person months per participant:	6	12	4	8	7	
Start month	6	**End month**		36		
Objectives						

WP4 will combine all the parts provided by WP1, WP2, and WP3 to compose power system network models of different scales and complexity for analysis. The idea is to equip the toolbox with methods to provide different types of analytic studies, ranging from fast and efficient system linearization for eigenvalue analysis, to complex region of attraction analysis. The end goal is to design a toolbox to assist the end-users with decision-making processes by providing fast and efficient analytic results. However, the toolbox will also provide the means for more extensive and accurate analytic studies at different computation timescales. There will be a trade-off between computational time and having accurate "enough" or rather suboptimal solutions, which the toolbox will make clear to the end-user by providing the respective assumptions and confidence limits of the presented results.

- Develop a TenSyGrid Toolbox via incremental grid models and application in real-world grid examples; and explore different grid control designs.

Description of work

T4.1 Identification of Use-cases (**UPC**, eRoots, UM, HAW, IWES) M6-M12: This task will conduct a thorough analysis of the different possible use-cases. Workshops with several stakeholders (e.g. network operators, equipment manufacturers, etc.) will be organized to discuss the potential of the tools developed in WP1-WP3 aiming to identify which are the applications where the use of multilinear models will be of high interest. Based on the collected input, a roadmap for the TenSyGrid Toolbox will be completed, detailing the key use cases identified and detailing the key relevant tools to be further developed in Task 4.3.

Distribution of work: UPC will lead the organization of the discussions with operators. HAW and IWES will contribute to the definition of the concept. eRoots and UM will support the redefinition of the applications.

T4.2 Multilinear Tools for Power System Stability Analysis (UPC, eRoots, HAW, IWES) M12-M24. This task will bridge the gap between the mathematical MTI tools developed in WP1 and WP2 and their application to a power systems environment. Starting with the relevant use-cases and applications identified in T4.1, this task will establish a clear link between the applications (e.g., real-time direct stability assessment in the large-signal regime) and the MTI tools to be used (e.g., MTI-based computation of the region of attraction). Once these links have been defined, the task will implement each of the key identified applications, adjusting the necessary methods on the mathematical side to ensure adequate performance.

Distribution of work: UPC, IWES and HAW will collaborate to bridge the gap between theory and practice, eRoots will support the power system applied view.

T4.3 Toolbox Framework Development and Implementation (**UPC**, eRoots, UM, HAW, IWES) M10-M27: This task will develop the TenSyGrid Toolbox framework to integrate the work outcomes from WP1-WP3. Such framework should consider the following functionalities: 1) a complete library of modular and scalable multilinear models of power system components (developed in WP4); 2) a network model construction tool capable of interconnecting the developed models, to speed up the complete system models; 3) a suite of tools for stability analysis both for the small- and large-signal operational regime of power networks (e.g. eigenvalue analysis, computation of region of attractions, etc.). As a key functionality of the framework, given the tensor-based approach used during the modelling stage, the toolbox will offer efficient compression for high complexity systems. The toolbox will be implemented on a high-level, versatile, high performance and efficient programming language, such as Python. Simple synthetic-based network tests will be developed to verify the basic functionalities during the development.

Distribution of work: UPC and eRoots will conceptualize and develop the toolbox; IWES and HAW will support with the fundamental tools, UM will support with the application.

T4.4 Network Analysis Applications (**UPC**, eRoots, UM, HAW, IWES) M10-M36: This task will develop the key network analysis applications identified for the TenSyGrid Toolbox in WP1. The tool will leverage the multilinear-based methods capable of analyzing grid topologies of incremental complexity (WP1-WP2) as well as employ WP3 models to ease its usage in a real environment. First, a complete testing guideline for validation of the toolbox applications will be compiled. The validation will progressively increase in complexity, starting with application tests in simple but rather useful models (e.g. 2-node system, Kundur 2 area model and the IEEE 9 bus system) and advancing to medium-sized systems (e.g. IEEE 118 system) before tackling real network issues. For each considered model, two different scenarios will be considered, a more conventional synchronous generation-based system and a power electronics dominated one (e.g. high penetration of renewable energy), to validate the tool performance

in a wide variety of network states. Additionally, a list of synthetic issues to be tested in each system, considering both the small- and large signal regime, will be compiled in the guideline to secure the adequate functioning of the tool selected applications. Once the guideline has been completed the TenSyGrid Toolbox will be tested alongside the construction of each of the models discussed above, progressively increasing in complexity to facilitate the troubleshooting and results interpretation. Finally, the toolbox outcomes will be discussed with the participants of T4.1 discussion forum to capture their feedback and use it for implementing a continuous improvement.

Distribution of work: UPC and eRoots will implement the network analysis applications; IWES and HAW will support basic tools to deploy the applications, UM will bridge the applications developed with WP5.

Deliverables (PU: Public, CO: Confidential)

D4.1 TenSyGrid Toolbox Application Roadmap (UPC, PU, M24): Report detailing the key use-cases and applications of the toolbox.

D4.2 TenSyGrid Toolbox Framework (UPC, PU, M27): Report including a detailed description of the framework developed, detailing the structure, functionalities, and code.

D4.3 TenSyGrid Toolbox Applications (UPC, PU, M36): Report validating the applications and basic tests of the TenSyGrid Toolbox version 1.0.

Milestones

MS4.1 Completed TenSyGrid Toolbox Roadmap (UPC, M16): Deliverable D4.1 includes a detailed report on the key use-cases and applications of the toolbox.
MS4.2 Complete TenSyGrid Toolbox (UPC, M24): The Toolbox will be validated, tested in different scenarios (reported in D4.3) and uploaded into a public repository (e.g. GitHub) together with its documentation and worked examples.

WP5 Demonstration and Commercial Feasibility (Lead: UM)

Work package number	5	**Lead beneficiary**			UM	
Work package title	Demonstration and Commercial Feasibility					
Participant number	1	2	3	4	5		
Short name of participant	IWES	UPC	HAW	eRoots	UM		
Person months per participant:	2	4	1	6	18		
Start month	8	**End month**		36			

Objectives

- Explore TenSyGrid Toolbox compatibility with commercial software in accordance to SO5
- Test TenSyGrid Toolbox functionality via incremental grid models and application in real-world grid examples

Description of work

T5.1 TenSyGrid Toolbox Commercial Software Integration (**eRoots**, UM, UPC, IWES) M8-M30: The software interfaces to commercial solutions, primarily, IPSA, need to be identified to connect the TenSyGrid Toolbox, following SO5. IPSA provides an interface for data exchange via Python, which needs to be evaluated if it suffices the requirements for the TenSyGrid Toolbox. Other approaches will be explored if necessary. A parser to handle file formats from other commercial tools such as PSSE, PowerFactory or the Common Information Model will be built.

Distribution of work: eRoots leads the development of the interface of TenSyGrid toolbox with support from UPC. UM provides the expertise as a software user of IPSA.

Task 5.2: Mid-Size Network Model (**UM**, eRoots, UPC, IWES, HAW) M18-M26: This task will develop a model of a mid-sized network model such as the CIGRE LV Distribution Network Benchmark for the Integration of Renewable and Distributed Energy Resources or IEEE 118 bus model. This model shall be developed in the IPSA software package and also in the MATLAB/Simulink with TenSyGrid toolbox developed in WP4. The effectiveness of the analytical tool will be evaluated and compared for proof of concept its ability to capture relevant non-linear dynamics (SO1).

Distribution of work: UM develops the CIGRE LV Network Model in IPSA as well as using TenSyGrid toolbox. eRoots and UPC will help with the use of the TenSyGrid toolbox to model the LV Network Model in MATLAB. IWES and HAW will offer their support during the evaluation and comparison of the results obtained.

Task 5.3: Large-Scale Network Model (**UM**, eRoots, UPC, IWES, HAW) M24-M30: Similar to Task 5.2, a large-scale power system model (e.g.: ACTIVSg2000 bus network or similar) will be modelled in a power system analysis software firstly, and subsequently re-developed using the TenSyGrid Toolbox. The effectiveness of the analytical tool will be evaluated and compared. This task is primarily designed to show the strength of the modelling framework to scale a mid-sized to a large-scale network model as outlined in SO4. Similar to Task 5.2, this task is aimed at addressing SO1 on a larger power system model.

Distribution of work: UM will develop a large-scale Power Network Model both in a commercially available software and also using TenSyGrid toolbox. eRoots and UPC will help with the use of the TenSyGrid toolbox to model the LV Network Model in Matlab. IWES and HAW will offer their support during the evaluation and comparison of the results obtained.

Task 5.4: Maltese Electrical Network Simulation Case-study (**UM**, eRoots, UPC, IWES, HAW) M24-M36: The TenSyGrid Toolbox will be used to model and analyze the Maltese Electrical Network or part thereof to validate the performance of the developed models under different scenarios. This will consist of linear simulations such as load flow analysis at critical operating points and non-linear simulations such a transient stability analysis during faults (e.g.: loss of a 200MW interconnection to Sicily, sudden loss of PV generation, etc...) A comparison of the results on the effect of having interpretable models through system identification vis-a-vis OEM black-box models will be carried out (SO3).

Distribution of work: UM will develop the Maltese Electrical Network Model in IPSA and in Matlab/Simulink using TenSyGrid toolbox. eRoots and UPC will help with the use of the TenSyGrid toolbox to model the Maltese Electrical Network Model in Matlab. IWES and HAW will offer their support during the evaluation and comparison of the results obtained.

Task 5.5: Malta Feasibility Study Towards 100% Renewables (**UM**, eRoots, UPC, IWES, HAW) M28-M36: This task will develop a simulation model assuming the planned offshore renewables installations. The study will consist of determining the maximum acceptable limit of renewables penetration in Malta without the need to run a number of simulation analysis to identify potential instability scenarios. This work is aimed to partly address SO2 where system operators can increase the limit of VRE penetration without reverting to conservative stability limits. Load flow and transient stability simulations at critical operating points using IPSA and another model using the TenSyGrid Toolbox will be carried out to highlight the improvement in stability assessment.

Distribution of work: UM will carry out technical feasibility studies using IPSA and the TenSyGrid toolbox. eRoots and UPC will support with the implementation and evaluation of the technical feasibility.

Deliverables (PU: Public, CO: Confidential)

D5.1 Evaluation of TenSyGrid Toolbox Functionalities (UM, PU, M30): Evaluation report validating the functionalities of the TenSyGrid Toolbox version 1.0 with LV Network Models

D5.2 Testing with Real-World Examples (UM, CO, M36): Evaluation report with a real-world example.

D5.3 Malta Feasibility Study Towards 100 % Renewables (UM, CO, M36): Evaluation report validating the functionalities of the TenSyGrid Toolbox version 1.0.

Milestones

MS5.1 Benchmarking of Toolset (UM, M30): TenSyGrid Toolbox benchmarked with existing software packages.
MS5.2 Improvement in Stability Assessment for a Real-world Case (UPC, M36): Compare the improvement in performance in assessing network stability of the Toolbox.

WP6 Reporting and Knowledge Community (Lead: IWES, Participants: UPC, UM, HAW, eRoots)

Work package number	6	**Lead beneficiary**			IWES
Work package title	Reporting and Knowledge Community				
Participant number	1	2	3	4	5	
Short name of participant	IWES	UPC	HAW	eRoots	UM	

Person months per participant:	1.5	1	1	1	1	
Start month	1			**End month**	36	

Objectives

- Periodic reporting on the project
- Participation in working groups
- Contribution to the knowledge community in the framework of CETP

Description of work

T6.1 Reporting (**IWES**, HAW, UPC, eRoots, UM) M1-M36**:** Creation of annual reports describing status of tasks, deviations, results, and milestone achieved, challenges faced and their mitigation. At the end of project, final report, and a public abstract with main results of the project.

T6.2 Formative Evaluation of Project (**IWES**, HAW, UPC, eRoots, UM) M1-M36**:** Participation in CETP working groups for the exchange of results and knowledge with other projects under the umbrella. Feedback to other funded projects with common synergies & vice versa. Annual uptake of feedback from CETP monitoring group. In-person feedback from Coordinators of other funded project at peer-to-peer online meetings where project results are presented and discussed. Then implementation of feedback including from own advisory board. Interactions with other projects including feedback help in the new modeling class and stability analysis tools development. Exchange with other experts helps to identify other use cases where the project outcomes could be useful.

T6.3 Contribution to Other Knowledge Community Activities (**IWES**, HAW, UPC, eRoots, UM) M1-M36**:** As a part of thematic working group of TRI, provide contribution to the living documents, contribute to continuous development of spotlights and policy briefs related to the working group's topics by clarifying conclusions, giving feedback and examples etc. from own and other project results. Represent project in the working group meetings -one onsite meeting and two online meetings per year. This includes preparation for, participation in and follow-up of the meetings. Co-create joint understanding of thematic and cross-cutting challenges by providing learnings from own project and from others. Detect synergies between funded project and take part in joint communication and dissemination activities via online meetings, workshops etc. For example, with UPCs other CETP TRI2 funded project WIND-DIGIPOWER and with eRoots and UPCs project FLEXDIS. Take part in CETP annual project events (hybrid).

Distribution of work in WP6: IWES prepares and submits the reports and public abstract after collecting input from all partners. IWES represents project in Coordinator events. All partners contribute to CETP working groups after internal discussions and take part in meetings. IWES prepares material for knowledge community based on contributions from partners. All partners take part in meetings and detect synergies with other projects.

Deliverables (PU: Public, CO: Confidential)

D6.1 First Annual Report (IWES, CO, M12): Project report at the end of 1st year.

D6.2: Second Annual Report (IWES, CO, M24): Project report at the end of 2nd year

D6.3: Final Report (IWES, CO, M36): Report describing project activities and outcome at the end of the project.

D6.4: Project Summary (IWES, PU, M36): A public abstract of project and outcome at the end of the project

WP7 Management and Dissemination (Lead: IWES)

Work package number	7	**Lead beneficiary**				IWES	
Work package title	Management and Dissemination						
Participant number	1	2	3	4	5		
Short name of participant	IWES	UPC	HAW	eRoots	UM		
Person months per participant:	2.5	1	1	1	1		
Start month	1	**End month**		36			

Objectives

- Coordinate the scientific work of the partners, monitor progress, check potential risks, and eventually implement mitigation actions, safeguard the quality of deliverables and reports.
- Set up administrative processes (for deliverables, reporting, etc.), organize meetings, support partners in administrative issues.
- Develop a comprehensive communication, dissemination, and exploitation strategy.

Description of work

T7.1 Project Management and Administration (**IWES**, UPC, HAW, eRoots, UM) M1-M36: The coordinator serves as central contact to the outside, is responsible for financial issues, checks and submits reports to funding agency. As primus inter pares, IWES coordinates the scientific work, monitors the progress, and prepares the decisions in consultation with partners. Project risks are initially determined, regularly checked and eventually mitigation actions are implemented. The advisory board will be informed on important achievements and consulted on a yearly base. For day-to day administration and questions by partners, IWES act as the project office supporting organization of project kick-off meeting, plenary meetings at 6 months intervals (4 online and 3 on-site). It organizes compilation of project reports, quality check of deliverables, milestones via internal review process and supervises contractual obligations.

T7.2 Data Management (**IWES**, UPC, HAW, eRoots, UM) M1-M36: Data management in the project follows the FAIR principles for data handling. Shortly after kick-off a platform for project-internal communication is set up at MS Teams/SharePoint (chat, calendar, Gantt-chart, data exchange) by IWES. Data that are connected to scientific publications or other public final data sets are made available in an open data format on a publicly accessible repository TBD. Final data sets considered relevant for further IP protection, will not be published but kept for exploitation.

T7.3 Results Dissemination, Communication, and Exploitation (**IWES**, UPC, HAW, eRoots, UM) M1-M36: A plan for the dissemination and communication of project results and activities is drafted in M6 and regularly updated by IWES with the support of the other partners. The work starts with a creation of visual identity of the project which includes, project logo, templates, webpage, social media -LinkedIn page and X account. After kick-off a press release by all partners on their respective sites and local media is planned. Project results, news at regular intervals displayed at the webpage. Further on results are presented in national and international workshops, conferences, stakeholder events, and open-access scientific articles in peer review journals. In line with the consortium- and the grant agreement the consortium intends to protect any commercially significant innovations and results by suitable forms (e.g., licenses). For findings requiring IPR protection, the exploitation measures are checked at the plenary meetings. A workshop on multilinear modeling will be organized. To aid exploitation a half-day seminar will be organized by UM to disseminate and discuss the project achievements with the local distribution system operator (Enemalta plc) and other stakeholders such as the regulator for energy and water serves (REWS) and the energy and water agency (EWA). The results of TenSyGrid will also be presented to the general public and external stakeholders in a non-technical form through talks, panel discussions in events, for example, The Science in the City – European Researchers' Night Festival in Malta.

Distribution of work for WP7: IWES coordinates, administrates, and manages TenSyGrid project with active support of all partners. IWES provides platform for internal communication and data storage. Project webpage content by IWES. All partners take part in dissemination, communication and exploitation of their respective results and overall project achievements. Based on contribution of partners IWES prepares and submits PEDR. UM organizes the stakeholder seminar.

Deliverables (PU: Public, CO: Confidential)

D7.1 Project Webpage and Visual Identity (IWES, PU, M4): Internal communication channel, project webpage and logo.

D7.2 Plan for Exploitation and Dissemination of Results (IWES, CO, M6): Plan for dissemination of results, data use and exploitation strategy. The document will be updated annually and the PEDR activities carried out in project will be part of annual and final project report.

3.2 Risk analysis

Table 4: Critical risks of implementation and mitigation measures

Description of risk (indicate level of (i) probability, and (ii) severity: Low/Medium/High)	Work package(s) involved	Proposed preventive/mitigation measures
Dynamic properties not conservable during model decomposition (Medium, High)	WP1, WP2	Relax applied definitions of stability, controllability, and observability.
Hybrid MTI models struggle to capture some non-linearities during validation (Medium, Medium)	WP3	Work on possible relaxations on the modelling to allow applying the developed tools.
No formal general grid stability certificates possible (High, High)	WP4	Restrict operational ranges for more conservative certificates.
Inadequate representation of the grid transient behavior (Medium, High)	WP4	Restrict operational range of the models.
Framework development incompatibilities with existing software (Medium, High)	WP4	Rebuild/translate the tools to a common programming language (e.g. Python)
Implementation delays due lack of timely support from DSO (Low, Medium)	WP5	Limit the size of the Maltese power system
TenSyGrid toolbox integration problems with the planned commercial software package (High, High)	WP5	Explore possible simplifications and alternative software packages

3.3 Partners involved and work distribution

To achieve all the objectives set by TenSyGrid, an interdisciplinary approach is clearly required. To tackle this challenge, the consortium for this project was built with interdisciplinarity at its core and a vertical integration strategy to foster synergy. Therefore, each partner complements each other's expertise, promoting a close collaboration scheme to enable innovative solutions.

The expertise of Gerwald Lichtenberg's team at the HAW Hamburg in multilinear modelling is unique in Europe, particularly for applications like controller design and fault diagnosis. Furthermore, they have developed a toolbox for multilinear time-invariant system modelling and controller design (MTI Toolbox). Up to now, the HAW has applied multilinear modelling techniques to small or medium size HVAC systems in various projects. Today, the HAW focuses on extending the multilinear modelling framework to large-scale systems as well as deeper analysis methods, making them the ideal leading partner for WP1.

The AC/DC grids and electric power conversion group of the research and technology transfer center CITCEA-UPC (co-led by Eduardo Prieto-Araujo) has wide experience in the analysis of modern power systems dominated by renewable generation and power electronics, which they acquired through their participation in several competitive and industrial projects. Therefore, UPC can ensure that the multilinear models capture the relevant components and their behavior correctly in WP3, and that the appropriate analyses are performed in WP4.

eRoots is a spinoff company of CITCEA-UPC whose aim is to bring the ideas developed there to the market in the form of products. eRoots specializes in performing electrical studies for modern power grids, especially those with numerous converters. One of the other core activities of eRoots is the development of software for grid operators, that is, tools that are both useful for steady-state and dynamic analysis. Therefore, eRoots provides the insights of the end-user perspective as well as how to develop tools for commercial software in WP5.

The team Integration of Local Energy Systems (ILES) at Fraunhofer IWES is headed by Jan Wenske with Gerwald Lichtenberg as scientific director (HAW). ILES research focus is the development of new modelling and control techniques to enable the integration of local energy systems to the grid. Their expertise makes them an ideal fit to bring the multilinear control concepts from WP1 into power system applications through WP4.

The Project team at University of Malta (UM) consists of the following three senior researchers: Dr. Ing. John Licari (PI), Prof. Cyril Spiteri Staines and Prof. Alexander Micallef. They are all members of the Department of Electrical Engineering. The academic staff are actively involved in state-of-the-art research in: Power

Systems, Microgrids and Smart Grids; Power Electronic Converters; Renewable Energy Systems for Grid Integration; Energy Storage and Electric Transportation. Their expertise in modelling of power electronics and power systems, and their knowledge on the Maltese Power System will ensure the successful completion of WP5. UM replaces TU Delft, who had to step-out after the first stage of proposal evaluation due to funding constraints by their national agency for this call module. Nevertheless, Kim Bastelier, co-director of Delft Tensor AI Lab, Delft University of Technology supports the consortium as project advisor.

3.4 Previous experience

Activities/research/publication	Partner	Relevance to the TenSyGrid
G. Lichtenberg, G. Pangalos, C. Cateriano Yáñez et al., "Implicit multilinear modeling: An Introduction with Application to Energy Systems," at - Automatisierungstechnik, vol. 70, no. 1, 2022.	HAW, IWES	Introduction of the implicit multilinear modelling class with application to an electrolyzer example
C. Kaufmann, Crespí de Valldaura Garcia, Diego, G. Lichtenberg et al., "Efficient Linearization of Explicit Multilinear Systems using Normalized Decomposed Tensors," IFAC-PapersOnLine, vol. 56, issue 2, 2023, pp. 7312-7317.	IWES, HAW	Efficient linearization method enabling fast eigenvalue analysis of multilinear models
Project AGISTIN, Horizon Europe, 14 partners, 8.8 Mio. €, 2023-2026	IWES	Model-based control using multilinear models for local energy systems
Project "Norddeutsches Reallabor", BMWi, 40 partners, funding 52 Mio €, 2021-2025	IWES	Modeling of power networks and microgrids and grid supporting control development
Project SONDE Program FHProfUnt, BMBF, 3 partners, 1 Mio €, 2020-2024	HAW	Project on Fault Diagnosis of building systems based on multilinear models
Project OBSERVE, BMWi, 5 partners, 2.3 Mio €, 2014-2018	HAW	Project on Controller Design and Fault diagnosis of HVAC Systems based on multilinear modeling
J. Licari, C. Spiteri Staines, A. Micallef, S. Hopert, "Experimental Validation of a Low-Voltage Regulator Model - A Case Study of a Maltese Low-Voltage Feeder" IET Renewable Power Generation, Vol 18, Issue 4, p. 679-689, Jan 2024. Available: Link	UM	Analysis of high penetration of RES on the Maltese LV Power Network
PROJECT: ESTELLE (2021) funded through the EWA fund 2021	UM	Energy Storage and Power-to-X technologies for optimal integration of renewable generation
PROJECT: REVOLT (2022) funded through the EWA fund 2022	UM	Reducing RES Energy generated intermittency through Voltage Regulation
Tools for modern power systems, granted by ACCIÓ (Catalan Government)	eRoots	Development of algorithms to study the effects of renewables in modern grids, especially focused on short-circuits
C. Collados-Rodriguez, M. Cheah-Mane, E. Prieto-Araujo and O. Gomis-Bellmunt, "Stability Analysis of Systems With High VSC Penetration: Where Is the Limit?," in IEEE Transactions on Power Delivery, vol. 35, no. 4, pp. 2021-2031, Aug. 2020.	UPC	Identification of potential stability limits in power systems with a high penetration of power electronics
Carlos Collados-Rodriguez, Marc Cheah-Mane, Eduardo Prieto-Araujo, Oriol Gomis-Bellmunt, Stability and operation limits of power systems with high penetration of power electronics, International Journal of Electrical Power & Energy Systems, Volume 138, 2022, ISSN 0142-0615,	UPC	Development of a indices-based methodology to assess instabilities at power flow, small-signal and large-signal regime for modern power networks.
Cheah-Mane, M., Egea-Alvarez, A., Prieto-Araujo, E., Mehrjerdi, H., Gomis-Bellmunt, O., & Xu, L. (2023). Modeling and analysis approaches for	UPC	Compilation of methods for small-signal analysis in modern power electronics

small-signal stability assessment of power-electronic-dominated systems. Wiley Interdisciplinary Reviews: Energy and Environment, 12(1), e453.		dominated networks
Project: TED2021-130351B-C21: High-Precision High-Performance Computer-enabled Digital Twins for modern power system applications (HP2C-DT) (together with BSC)	UPC	Development of tools for modern power system analysis using the supercomputer enabled analysis tools
Project: Removing technical barriers of high renewable energy and power electronics dominated networks using grid-forming-based STATCOM devices and HVDC links (REFORMING -PID2021-127788OA-I00)	UPC	Assessing the impact of penetration of grid-forming in the power system using small-signal analysis and EMT simulation tools

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